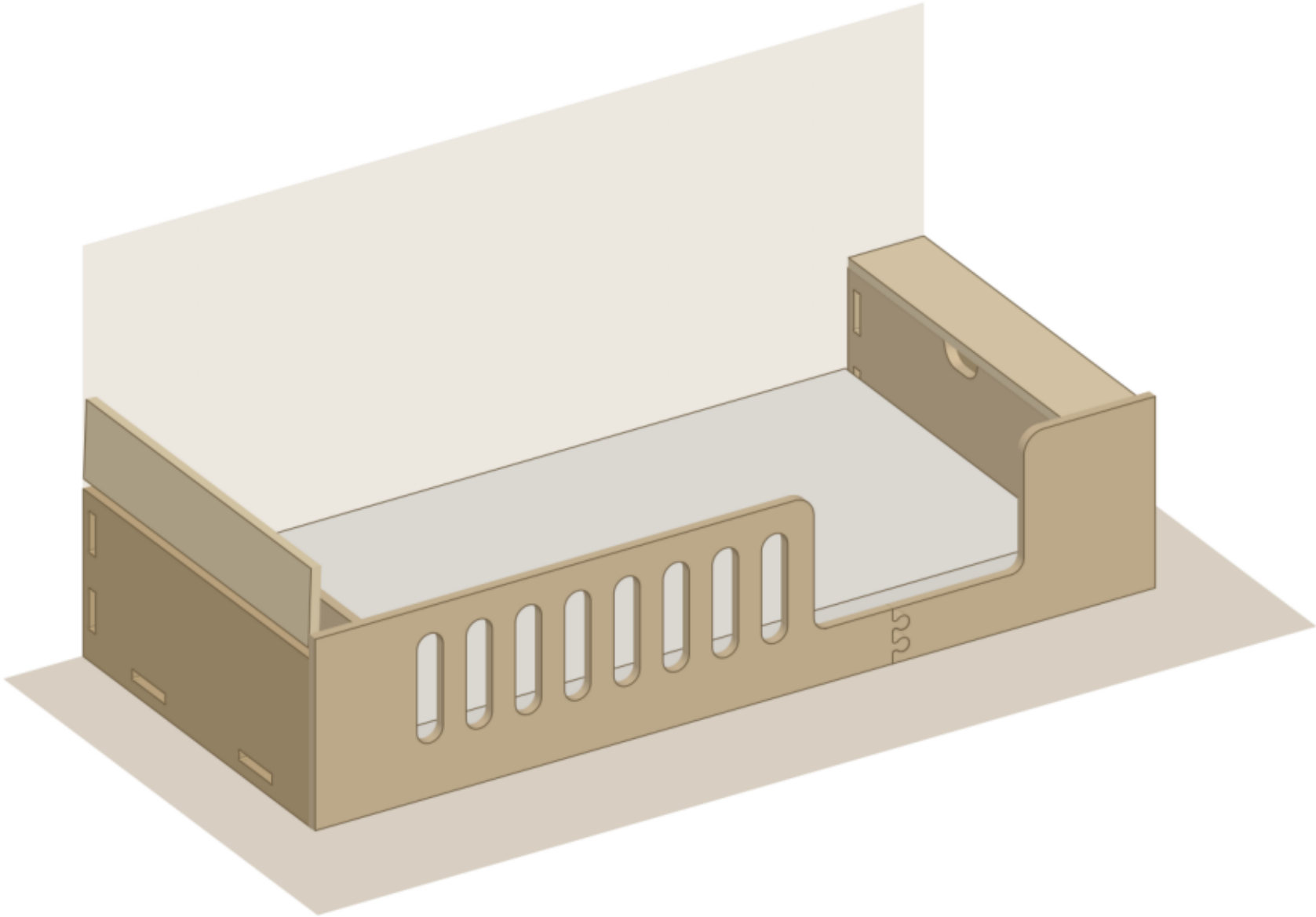
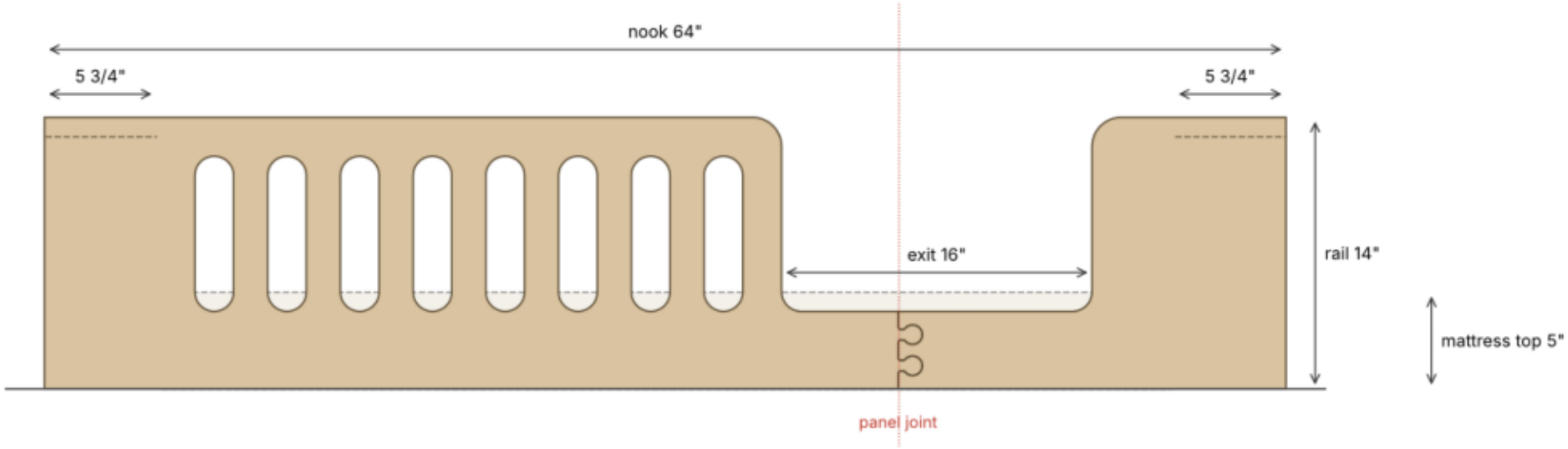
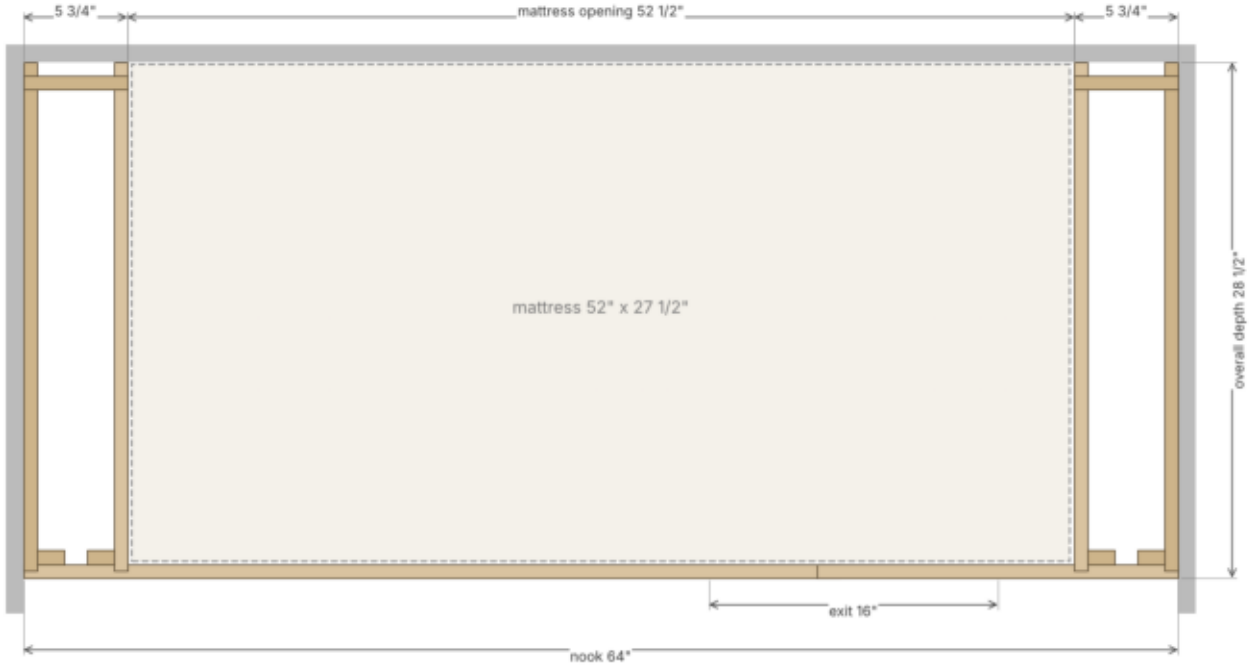
Floor bed, side exit



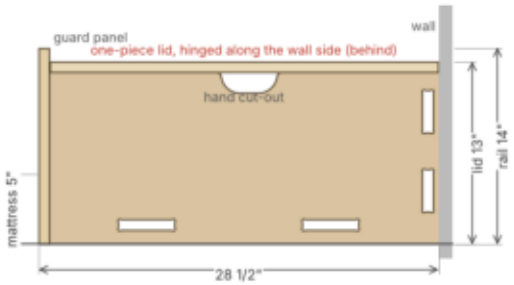
Floor bed, side exit: front view



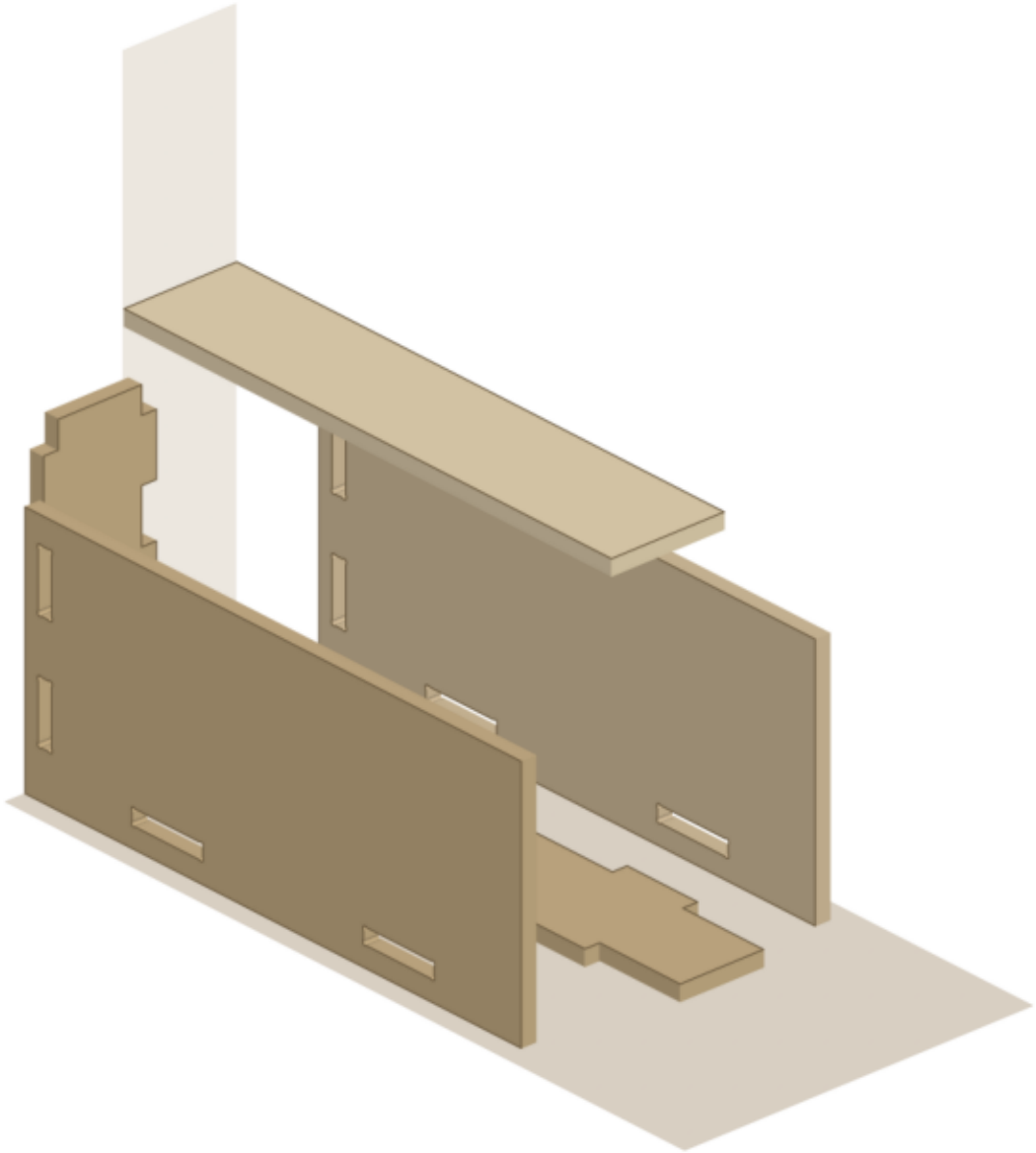
Plan (front of bed at the bottom)



Head box seen from the mattress

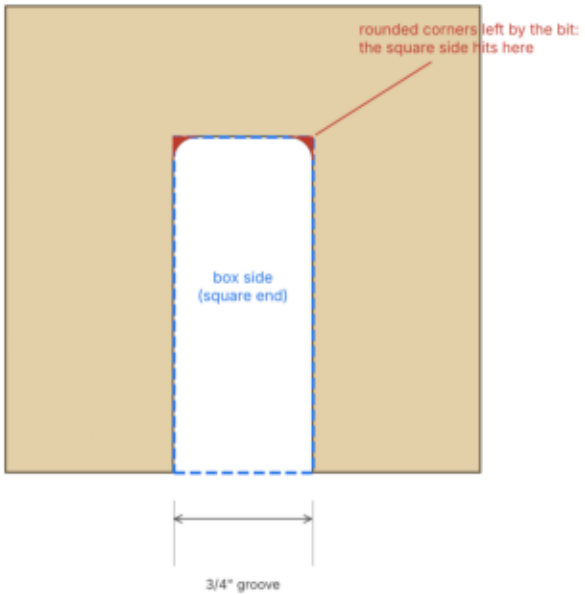


Head box, exploded (tabs pass through the sides)

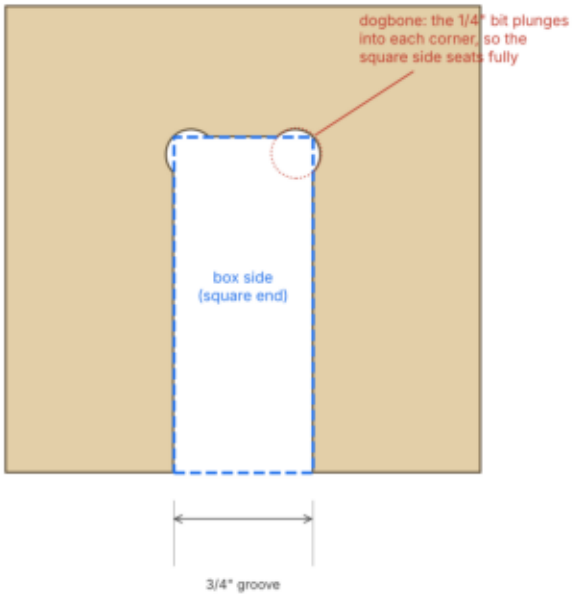


Dogbones: how a square part fits a CNC-cut groove (looking at the back of the front panel)

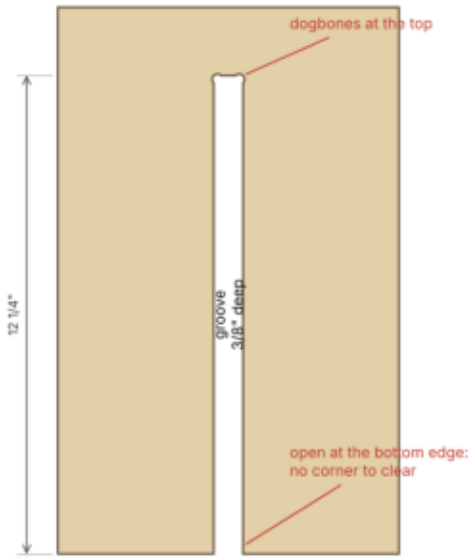
Without dogbones (red = wood that blocks the square corner)



With dogbones

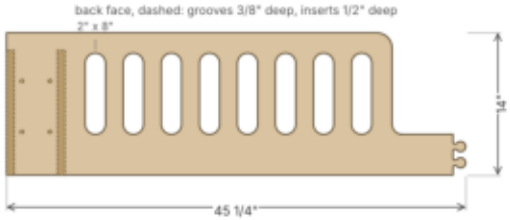


Back of the front panel, one groove

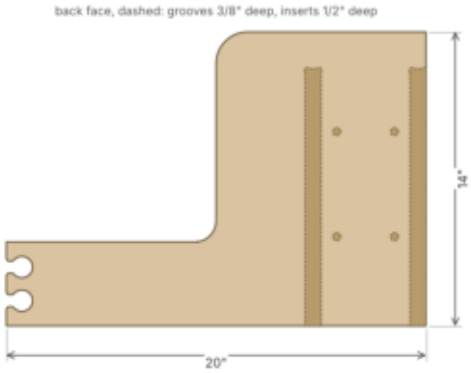


Floor bed, side exit: part drawings (inches, 3/4" plywood)

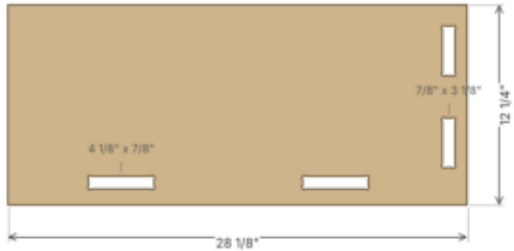
Front panel A (qty 1)
knob half



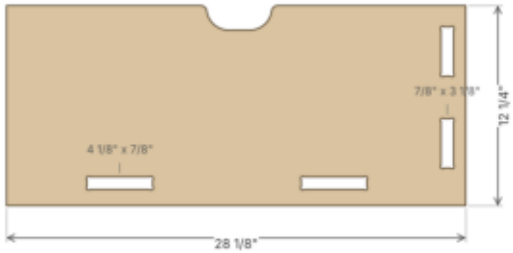
Front panel B (qty 1)
socket half



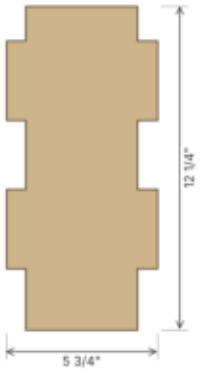
Box side (wall) (qty 2)
against the nook wall, 1 per box



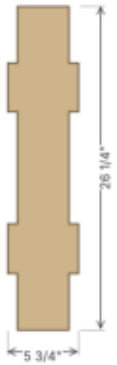
Box side (mattress) (qty 2)
faces the mattress; hand cut-out to open the lid



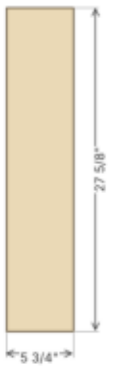
Box back (qty 2)
screw through into wall studs



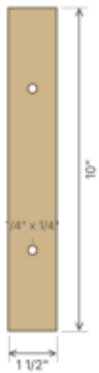
Box floor (qty 2)
sits 1" off the floor



Box lid (qty 2)
piano hinge along the long wall-side edge

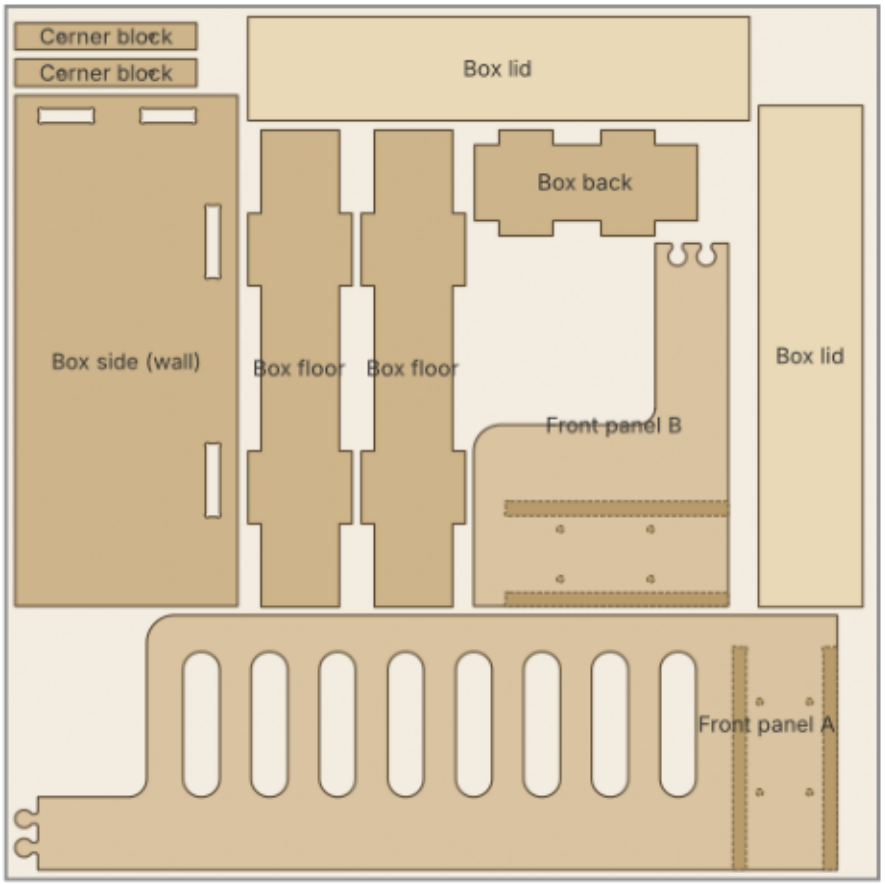


Corner block (qty 4)
glue + 2 pocket screws to the side; 2 bolts into the front panel

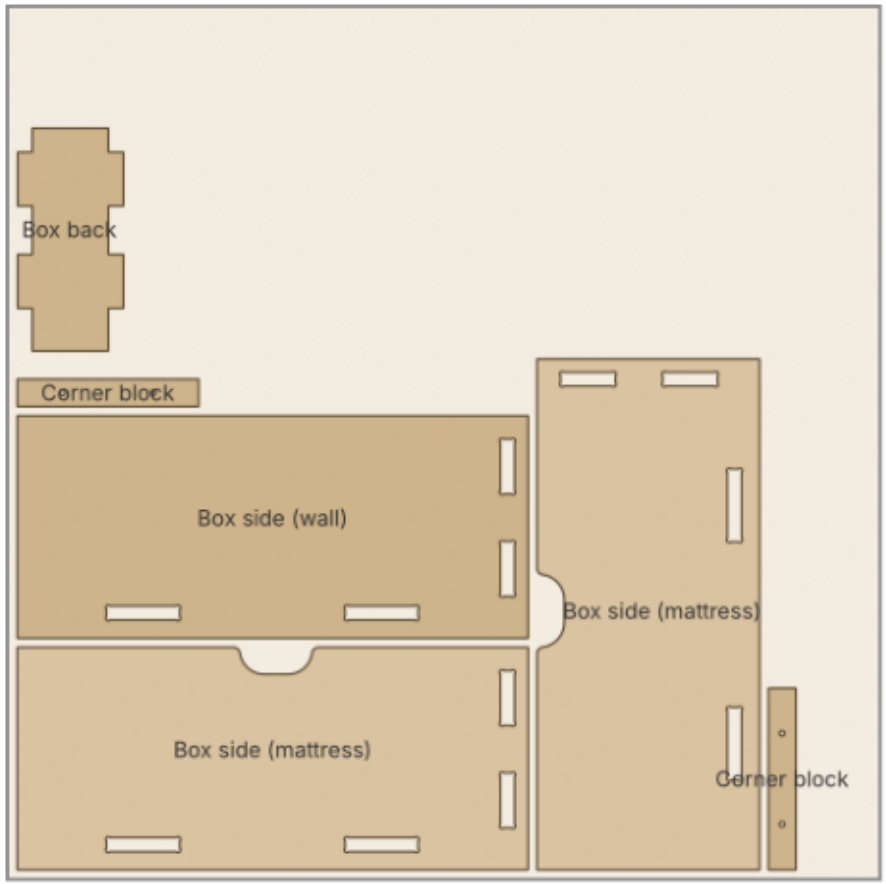


Floor bed, side exit: nesting on 48" square panels

4'x4' panel 1



4'x4' panel 2



Floor bed, side exit: cut list, hardware, assembly

Cut list

Part	Qty	Size (W x H)	Notes
Front panel A	1	45 1/4" x 14"	knob half
Front panel B	1	20" x 14"	socket half
Box side (wall)	2	28 1/8" x 12 1/4"	against the nook wall, 1 per box
Box side (mattress)	2	28 1/8" x 12 1/4"	faces the mattress; hand cut-out to open the lid
Box back	2	5 3/4" x 12 1/4"	screw through into wall studs
Box floor	2	5 3/4" x 26 1/4"	sits 1" off the floor
Box lid	2	5 3/4" x 27 5/8"	piano hinge along the long wall-side edge
Corner block	4	1 1/2" x 10"	glue + 2 pocket screws to the side; 2 bolts into the front panel

Key dimensions

Panels: 2 pieces of 4'x4' (1 x 4'x8' sheet(s) of 3/4" plywood, each cut in half)
Heights from floor: mattress top 5", rail top 14", box lids 13", exit sill 4"
Exit opening: 38" to 54" from the head wall; panel halves join at 44"
Head/foot boxes: 5 3/4" wide outside, 4 1/4" inside, 27 3/4" deep

Hardware

- Wood glue (PVA)
- 1-1/4" wood screws x 16 (box joinery)
- 1-1/4" pocket-hole screws x 8 (corner blocks to box sides)
- 1/4-20 threaded inserts for wood x 8, no longer than 3/8" (front panel)
- 1/4-20 x 1" bolts + washers x 8 (corner blocks to front panel)
- 2-1/2" screws x 4 into studs (box backs)
- Piano hinge 1-1/2" open width, 2 x 24" lengths (1 per lid)
- Soft-close lid stays x 2 (light-duty toy-box type)
- Shims as needed

Assembly

1. Cut all parts (sheets/panel_*.dxf). The two front-panel pieces are mirrored in the files: cut them BACK FACE UP so the POCKET layers (box-side grooves 3/8" deep, insert holes 1/2" deep) land on the back. Profile CUT_INSIDE before CUT_OUTSIDE, use tabs/onion skin on small parts, and dry-fit the puzzle joint and one groove on scrap first; adjust knob_clearance / tab_clearance if needed.
2. Ease every exposed edge with a 1/4" roundover (rail top, slots, exit opening, lid edges and the hand cut-outs). Leave joinery edges and faces that meet the wall square.